

Agentic AI Interview Questions with Answers

Q1. What is Agentic AI?

Agentic AI refers to AI systems that can autonomously make decisions, plan actions, use tools, and complete tasks with minimal human intervention.

Q2. How is an AI agent different from a chatbot?

A chatbot mainly responds to prompts, while an AI agent can reason, plan, remember, and take actions using external tools.

Q3. What are the core components of an AI agent?

Core components include an LLM, memory, tools/APIs, planning system, and execution loop.

Q4. Explain the ReAct framework.

ReAct combines reasoning and acting, allowing models to think step-by-step before taking actions.

Q5. What is tool calling in LLMs?

Tool calling allows LLMs to use external tools such as APIs, databases, or calculators.

Q6. What is function calling?

Function calling is a structured way for models to request execution of predefined functions.

Q7. Difference between workflow automation and autonomous agents?

Workflow automation follows fixed rules, while autonomous agents adapt and make decisions dynamically.

Q8. What makes an agent autonomous?

An agent is autonomous when it can independently decide actions and execute tasks toward goals.

Q9. What are planning and reasoning in AI agents?

Planning breaks tasks into steps, while reasoning helps the agent evaluate and decide the best actions.

Q10. Explain memory in AI agents.

Memory allows agents to store context, previous interactions, and learned information for future tasks.

Q11. What is RAG?

RAG stands for Retrieval-Augmented Generation, combining retrieval systems with LLMs for accurate responses.

Q12. Why use embeddings?

Embeddings convert text into vectors so semantic meaning can be searched and compared mathematically.

Q13. Difference between semantic search and keyword search?

Semantic search finds meaning-based matches, while keyword search relies on exact words.

Q14. Explain chunking strategies.

Chunking splits documents into smaller pieces for better retrieval and context management.

Q15. What causes hallucinations?

Hallucinations occur when models generate incorrect or fabricated information.

Q16. What is prompt engineering?

Prompt engineering is the process of designing prompts to improve model outputs.

Q17. What are system prompts?

System prompts define the AI's behavior, rules, and personality.

Q18. Explain structured outputs.

Structured outputs return responses in formats like JSON for reliability and automation.

Q19. What is prompt injection?

Prompt injection is an attack where malicious input manipulates model behavior.

Q20. How do you defend against prompt injection attacks?

Defenses include input validation, guardrails, permission controls, and separating instructions from user input.

Agentic AI Interview Questions & Answers (21-50)

Q21. What is hybrid search?

Hybrid search combines semantic vector search with traditional keyword search for better retrieval accuracy.

Q22. Explain cosine similarity.

Cosine similarity measures how similar two vectors are based on the angle between them.

Q23. How would you optimize retrieval latency?

Retrieval latency can be improved using indexing, caching, optimized chunking, and efficient vector databases.

Q24. Difference between GPT-4 and smaller models?

GPT-4 is more capable in reasoning and accuracy, while smaller models are faster and cheaper.

Q25. What is temperature in LLMs?

Temperature controls randomness in LLM outputs. Lower values are more deterministic.

Q26. Explain tokenization.

Tokenization splits text into smaller units called tokens for model processing.

Q27. What are context windows?

Context windows define how much information an LLM can process at once.

Q28. What are embeddings?

Embeddings are vector representations of data used for semantic understanding.

Q29. What causes hallucinations?

Hallucinations happen when a model generates false or fabricated information.

Q30. What is prompt engineering?

Prompt engineering is designing prompts to improve model responses.

Q31. What are system prompts?

System prompts define the AI's role, behavior, and restrictions.

Q32. What is chain-of-thought reasoning?

Chain-of-thought reasoning allows models to think step-by-step before answering.

Q33. Explain structured outputs.

Structured outputs provide responses in organized formats such as JSON.

Q34. Build an AI agent using Python.

An AI agent in Python can be built using an LLM, tools, memory, and a reasoning loop.

Q35. Write a tool-calling example using OpenAI SDK.

OpenAI SDK tool calling allows models to invoke predefined functions dynamically.

Q36. Create a weather agent.

A weather agent retrieves weather data from APIs and responds intelligently.

Q37. Implement memory for an AI chatbot.

Memory can be implemented using databases, vector stores, or conversation history.

Q38. Build a PDF Q&A; RAG system.

A PDF Q&A; RAG system extracts text, creates embeddings, stores vectors, and retrieves relevant chunks.

Q39. Create an AI email assistant.

An AI email assistant can summarize, draft, categorize, and automate email workflows.

Q40. Build an AI coding assistant.

AI coding assistants use LLMs to generate, debug, and explain code.

Q41. Create an agent that searches the web and summarizes results.

Web-search agents use search APIs and summarization pipelines to provide concise answers.

Q42. Design a multi-step reasoning agent.

Multi-step reasoning agents break tasks into smaller logical actions before execution.

Q43. Build a Slack or WhatsApp AI assistant.

Slack or WhatsApp assistants integrate APIs, messaging platforms, and LLM backends.

Q44. What is LangChain?

LangChain is a framework for building LLM-powered applications and agents.

Q45. Difference between LangChain and direct OpenAI API usage?

LangChain provides abstractions for chains and tools, while direct API usage offers more control.

Q46. What is LangGraph?

LangGraph is a framework for stateful and multi-agent workflows.

Q47. Explain agent executors.

Agent executors manage reasoning loops, tool usage, and task execution.

Q48. What are tools in LangChain?

Tools in LangChain are external functions or APIs available to agents.

Q49. What is a retriever?

A retriever fetches relevant documents or context for RAG systems.

Q50. Explain chains vs agents.

Chains follow predefined flows, while agents dynamically decide actions.

Agentic AI Interview Questions & Answers (50-100)

Q51. What are callbacks and tracing?

Callbacks and tracing help monitor and debug agent execution flows.

Q52. What is MCP (Model Context Protocol)?

MCP is a protocol that standardizes context sharing between models and tools.

Q53. Have you used OpenAI Agents SDK?

The OpenAI Agents SDK helps developers build autonomous AI workflows.

Q54. Design ChatGPT from scratch.

Designing ChatGPT requires LLMs, orchestration layers, memory, APIs, and scalable infrastructure.

Q55. Design a scalable AI assistant architecture.

Scalable AI assistants use distributed systems, caching, queues, and vector databases.

Q56. How would you reduce AI inference cost?

Inference costs can be reduced using caching, smaller models, batching, and optimized prompts.

Q57. How would you cache LLM responses?

LLM responses can be cached using Redis or semantic caching systems.

Q58. How would you handle rate limits?

Rate limits are handled using retries, queues, exponential backoff, and throttling.

Q59. How would you stream AI responses?

Streaming uses token-by-token generation for real-time responses.

Q60. How would you secure tool calling?

Tool calling security includes permission controls, sandboxing, and validation.

Q61. How would you monitor agent failures?

Agent failures are monitored using logs, traces, metrics, and alert systems.

Q62. How would you evaluate agent performance?

Performance is evaluated using benchmarks, human reviews, and automated metrics.

Q63. How would you deploy AI agents to production?

Production deployment uses APIs, containers, CI/CD pipelines, and monitoring.

Q64. What is prompt injection?

Prompt injection manipulates model instructions through malicious inputs.

Q65. How do you defend against prompt injection attacks?

Defenses include sanitization, role separation, and access controls.

Q66. Explain jailbreaking.

Jailbreaking bypasses AI safety restrictions using crafted prompts.

Q67. What are AI safety risks?

AI safety risks include misinformation, harmful actions, and data leaks.

Q68. How would you validate tool outputs?

Tool outputs are validated using schemas, constraints, and verification logic.

Q69. How do you restrict dangerous actions?

Dangerous actions are restricted through permissions and approval systems.

Q70. Explain human-in-the-loop systems.

Human-in-the-loop systems require humans to approve sensitive actions.

Q71. What are guardrails?

Guardrails enforce safety rules and behavioral boundaries.

Q72. How would you detect hallucinations?

Hallucinations are detected using verification systems and retrieval checks.

Q73. How would you handle sensitive data in RAG?

Sensitive data in RAG should be encrypted, filtered, and access-controlled.

Q74. What are multi-agent orchestration systems?

Multi-agent orchestration systems coordinate multiple specialized agents.

Q75. Explain reflection in AI agents.

Reflection allows agents to review and improve their own outputs.

Q76. What is tree-of-thought reasoning?

Tree-of-thought reasoning explores multiple reasoning paths before choosing one.

Q77. What is self-healing in agents?

Self-healing agents recover from errors and retry failed operations.

Q78. Explain planning algorithms for agents.

Planning algorithms help agents sequence tasks toward goals.

Q79. What is agent benchmarking?

Agent benchmarking measures accuracy, efficiency, and reliability.

Q80. How do autonomous coding agents work?

Autonomous coding agents analyze requirements, generate code, and debug automatically.

Q81. Explain computer-use agents.

Computer-use agents interact with GUIs, operating systems, and applications.

Q82. How do browser agents work?

Browser agents navigate websites, click elements, and extract information.

Q83. What are the limitations of current AI agents?

Current AI agents struggle with reliability, memory, reasoning, and safety.

Q84. An AI agent keeps looping infinitely. How do you debug it?

Infinite loops are debugged using logs, execution tracing, and step limits.

Q85. Your RAG system gives irrelevant answers. What would you improve?

Improve RAG with better chunking, embeddings, reranking, and retrieval tuning.

Q86. Users complain about hallucinations. What steps do you take?

Reduce hallucinations using RAG, validation, grounding, and better prompts.

Q87. Your vector DB grows to 100M embeddings. What changes?

Large vector DBs require sharding, indexing, distributed storage, and compression.

Q88. API costs become too high. How do you optimize?

Optimize API costs using caching, smaller models, batching, and prompt optimization.

Q89. Your agent chooses the wrong tools frequently. Fix it.

Wrong tool selection can be improved using better prompts and routing logic.

Q90. How would you build an AI SDR/sales outreach agent?

AI SDR agents automate lead discovery, messaging, and CRM integration.

Q91. How would you build a TikTok comment automation agent?

TikTok automation agents monitor comments, detect keywords, and send replies.

Q92. How would you build an AI research assistant?

AI research assistants retrieve, summarize, organize, and cite information.

Q93. What would you improve in current AI agents?

Current agents need better long-term memory, reasoning, and reliability.

Q94. What AI stack does the company use?

Companies may use OpenAI, LangChain, vector DBs, and cloud infrastructure.

Q95. Are agents rule-based or autonomous?

Agents can be fully autonomous or partially rule-based depending on use cases.

Q96. What vector DB infrastructure do you use?

Common vector DBs include Qdrant, Pinecone, Weaviate, and Chroma.

Q97. How do you evaluate agent quality?

Agent quality is evaluated through tests, benchmarks, and user feedback.

Q98. Do engineers own prompts in production?

Some companies allow engineers to manage prompts directly in production.

Q99. How do you handle AI safety?

AI safety uses moderation, guardrails, monitoring, and access restrictions.

Q100. Are you using fine-tuning or RAG?

Companies often combine fine-tuning with RAG for better performance.

Q101. What's the biggest challenge in your AI systems?

Major challenges include hallucinations, scalability, latency, and safety.